# Lesson 9.1 - Z-Scores, Confidence Intervals, and Margin of Error

> **Video:** How To Find The Z Score, Confidence Interval, and Margin of Error for a Population Mean by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

## Confidence interval for $\mu$ (when $\sigma$ is known)

$$\bar{x} - z\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z\frac{\sigma}{\sqrt{n}} \qquad \text{or} \qquad \bar{x} \pm \text{EBM}$$

**Margin of error**, also called the **error bound for the mean (EBM)**:

$$\text{EBM} = z\frac{\sigma}{\sqrt{n}}$$

## Finding the critical z-value

The area to the **left** of the positive critical value $z$ is

$$\text{left area} = \frac{\text{CL} + 1}{2}$$

where CL is the confidence level as a decimal. Look this area up in a z-table that gives **left** areas.

| Confidence level | Left area | $z$ |
|---|---|---|
| 90% | 0.95 | 1.645 |
| 95% | 0.975 | 1.96 |

For 90%, the table has 0.94950 at $z = 1.64$ and 0.95053 at $z = 1.65$. 0.95 is about halfway between them, so average: $z = 1.645$.

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## Example 1 - physics test, 95% CI

$\sigma = 5.4$, $n = 50$, $\bar{x} = 79$

1. $z$: left area $= \frac{\text{CL} + 1}{2} = \frac{0.95 + 1}{2} = 0.975 \rightarrow z = 1.96$
2. Standard error: $\frac{\sigma}{\sqrt{n}} = \frac{5.4}{\sqrt{50}} \approx 0.76368$
3. **(b) Margin of error:** $\text{EBM} = z\frac{\sigma}{\sqrt{n}} = 1.96 \times 0.76368 \approx 1.4968$
4. **(a) Interval:** $\bar{x} \pm \text{EBM} = 79 \pm 1.4968$

$$77.5032 < \mu < 80.4968$$

We are 95% confident that the mean score of the whole class lies between about **77.50 and 80.50**.

## Example 2 - chemistry test, 90% CI

$\sigma = 6.5$, $n = 100$, $\bar{x} = 82$

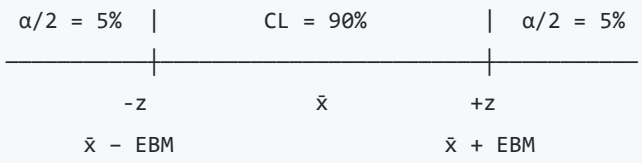
1. $z$: left area $= \dfrac{\text{CL}+1}{2} = \dfrac{0.90+1}{2} = 0.95 \rightarrow z = 1.645$
2. **(b) Margin of error:** $\text{EBM} = z\,\dfrac{\sigma}{\sqrt{n}} = 1.645 \times \dfrac{6.5}{\sqrt{100}} = 1.645 \times 0.65 = 1.06925$
3. **(a) Interval:** $\bar{x} \pm \text{EBM} = 82 \pm 1.06925$

$$80.93075 < \mu < 83.06925$$

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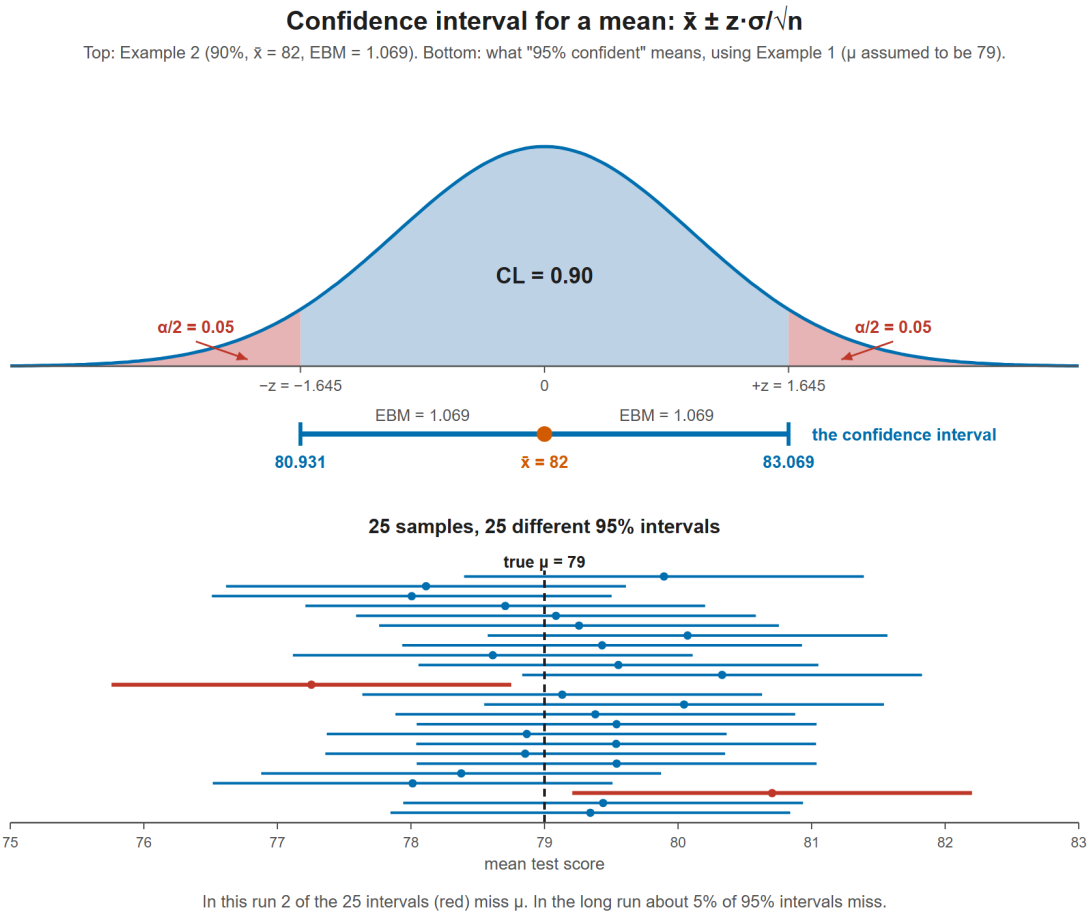
## Confidence interval vs confidence level

| Term | Meaning | Example |
|---|---|---|
| **Confidence interval (CI)** | The **range of values** $(\bar{x} - \text{EBM},\ \bar{x} + \text{EBM})$ | 80.93 to 83.07 |
| **Confidence level (CL)** | The **area under the curve** between $-z$ and $+z$ | 0.90 |
| $\alpha$ | The area **outside** the interval: $\alpha = 1 - \text{CL}$, split as $\alpha/2$ in each tail | 0.10 → 0.05 per tail |




### Interpreting "95% confidence"

If we took many samples and built a 95% interval from each one, about **95% of those intervals would contain the true mean** $\mu$, and about 5% would miss it. The video phrases this in terms of sample means landing in the interval; this is the more precise statement.




Top: the confidence level is the middle area between $-z$ and $+z$, and the interval reaches one EBM either side of $\bar{x}$. Bottom: 25 simulated samples give 25 different intervals. Most of them cover the true mean (dashed line), but a few miss it, which is what "95% confident" means in the long run.

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## What changes the margin of error?

$$\text{EBM} = z\,\frac{\sigma}{\sqrt{n}}$$

| Change | Effect on EBM | Why |
|---|---|---|
| **Higher** confidence level (90% → 95%) | **Larger** | $z$ grows, so the interval gets wider |
| **Larger** sample size $n$ | **Smaller** | $\sqrt{n}$ is in the denominator |

To get a smaller margin of error, **lower the confidence level** or **increase the sample size**.

The EBM is half the width of the confidence interval.

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## Key takeaways

- CI = $\bar{x} \pm z\,\sigma/\sqrt{n}$, and EBM = $z\,\sigma/\sqrt{n}$.
- Critical $z$: look up the left area $(\text{CL}+1)/2$. 90% → 1.645, 95% → 1.96.
- Higher confidence gives a wider interval. A bigger sample gives a narrower one.

# Lesson 9.2 - How to Find the Z-score Given the Confidence Intervals

> **Video:** How To Find The Z Score Given The Confidence Level of a Normal Distribution by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

## Common values to memorize

| Confidence level | $z$ |
|---|---|
| 90% | 1.645 |
| 95% | 1.96 |
| 98% | 2.33 |
| 99% | 2.575 |

## Method for any confidence level

**Step 1 - area to the left of $+z$:**

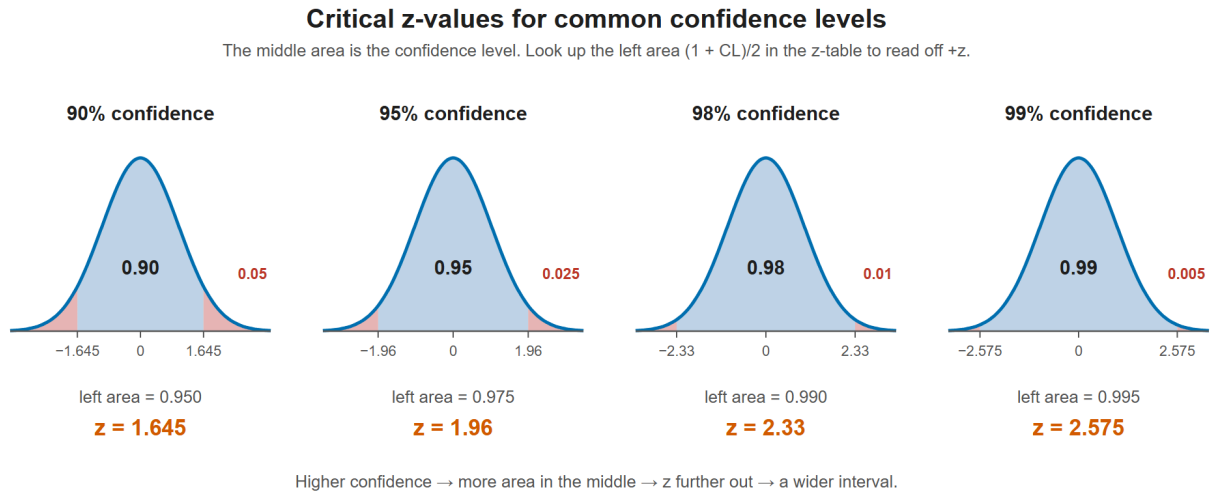
$$\text{left area} = \frac{1 + \text{CL}}{2}$$

**Why:** the confidence level is the area between $-z$ and $+z$. The leftover $1 - \text{CL}$ is split evenly between the two tails. For 95%:

```
 0.025  |          0.95           |  0.025
--------|-------------------------|--------
       -z           mean         +z
left area up to +z = 0.95 + 0.025 = 0.975
```

**Step 2 - find that area inside a positive z-table** (one that gives **left** areas). Read the row (ones and tenths digits) and the column (hundredths digit), then add them.

*Example:* 0.975 sits in row **1.9**, column **0.06**, so $z = 1.9 + 0.06 = 1.96$.



The confidence level is the middle area, and each tail gets half of what is left. The left area up to +z is (1 + CL)/2, which is the number to look up in the z-table.

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## Practice

**97% confidence**

$$\text{left area} = \frac{1 + \text{CL}}{2} = \frac{1 + 0.97}{2} = 0.985 \;\Rightarrow\; \text{row } 2.1,\ \text{column } 0.07 \;\Rightarrow\; z = 2.17$$

**92% confidence**

$$\text{left area} = \frac{1 + \text{CL}}{2} = \frac{1 + 0.92}{2} = 0.96$$

The closest table entry is 0.95994, in row **1.7**, column **0.05**, so $z = 1.75$.

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## Key takeaways

- Left area $= (1 + \text{CL})/2$. Look it up in the body of the z-table to read off $z$.
- If the exact area isn't in the table, use the closest entry, or average two entries that are equally close (e.g. 90% → 1.645).
- Know 1.645, 1.96, 2.33 and 2.575 by heart.

# Lesson 9.3 - Student's T-Distribution

> **Video:** Student's T Distribution - Confidence Intervals & Margin of Error by The Organic Chemistry Tutor
> **Chapter:** 9 - Confidence Intervals

## When to use t instead of z

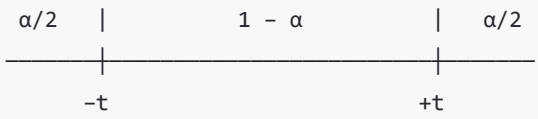
| Use the **normal (z)** distribution when... | Use **Student's t** when... |
|---|---|
| the **population** SD $\sigma$ is known | only the **sample** SD $s$ is known |
| $n \geq 30$ | $n < 30$ |

Having only $s$ and a small $n$ are both signs that you need $t$.

## Formulas

$$\bar{x} \pm t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}}, \qquad \text{EBM} = t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}}$$

| Symbol | Meaning |
|---|---|
| $\nu = n - 1$ | **Degrees of freedom** (Greek "nu") |
| $\alpha = 1 - \text{CL}$ | The total area outside the interval, i.e. both tails |
| $\alpha/2 = \dfrac{1 - \text{CL}}{2}$ | The **one-tail** area, used to look up $t$ |




**Reading a t-table:** go to the row for $\nu$ and the column for the one-tail area $\alpha/2$.

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## Example 1 - student weights, 95% CI

20 students: $\bar{x} = 165$ lb, $s = 4.5$ lb.

$s$ is a **sample** SD (the school has far more than 20 students) and $n = 20 < 30$, so use $t$.

- $\nu = n - 1 = 20 - 1 = 19$
- $\alpha/2 = (1 - \text{CL})/2 = (1 - 0.95)/2 = 0.025$
- Table: $t_{19,\,0.025} = 2.093$

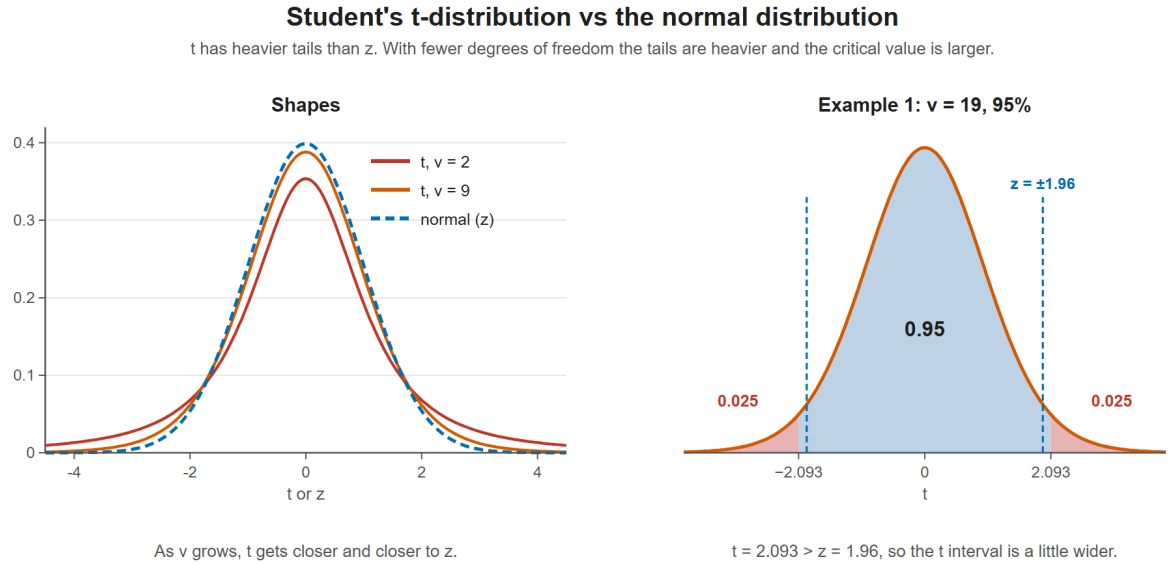
**(b) Margin of error**

$$\text{EBM} = t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}} = 2.093 \times \frac{4.5}{\sqrt{20}} = \frac{9.4185}{4.4721} \approx 2.106$$

**(a) Interval**

$$\bar{x} \pm \text{EBM} = 165 \pm 2.106 \;\Rightarrow\; 162.894 < \mu < 167.106$$

We are 95% confident that the mean weight of all the school's students is between about **162.9 and 167.1 lb**.




Left: t has heavier tails than the normal curve, especially with few degrees of freedom. Right: for $\nu = 19$ the 95% critical value is 2.093, a bit further out than z = 1.96, so t intervals are slightly wider.

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## Example 2 - chemistry scores, 90% CI

The class has 500 students. 10 scores were sampled: 76, 84, 69, 92, 58, 89, 73, 97, 85, 77.

**(a) Sample mean and SD**

$$\bar{x} = \frac{\sum x}{n} = \frac{800}{10} = 80$$

Distances from 80: 4, 4, 11, 12, 22, 9, 7, 17, 5, 3 (squaring removes the signs).

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} = \sqrt{\frac{4^2 + 4^2 + 11^2 + 12^2 + 22^2 + 9^2 + 7^2 + 17^2 + 5^2 + 3^2}{10 - 1}} = \sqrt{\frac{1234}{9}} \approx 11.709$$

**(b) Margin of error at 90%**

- $\nu = n - 1 = 10 - 1 = 9$
- $\alpha/2 = (1 - \text{CL})/2 = (1 - 0.90)/2 = 0.05$
- Table: $t_{9,\,0.05} = 1.8331$

$$\text{EBM} = t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}} = 1.8331 \times \frac{11.709}{\sqrt{10}} \approx 6.787$$

**(c) 90% confidence interval**

$$\bar{x} \pm \text{EBM} = 80 \pm 6.787 \;\Rightarrow\; 73.213 < \mu < 86.787$$

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## Key takeaways

- Use $t$ when $\sigma$ is unknown (you only have $s$), especially when $n < 30$.
- Degrees of freedom $\nu = n - 1$. Look up the one-tail area $\alpha/2 = (1 - \text{CL})/2$.

- CI = $\bar{x} \pm t\, s/\sqrt{n}$. The $t$ value is larger than $z$ for small samples, which gives a wider and more cautious interval.

# Lesson 9.4 - Finding a Confidence Interval of a Population Proportion

> **Video:** Finding The Confidence Interval of a Population Proportion Using The Normal Distribution by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

## The big idea

You want to know what **fraction** of a large group has some property, for example "what percent of students at this university own a laptop?" Asking everyone (maybe 10,000 people) is too much work, so you ask a **sample** instead.

The sample gives you **one estimate**. Because it's only a sample, that estimate is probably a little off. A confidence interval turns the single estimate into a **range** and says how confident you are that the true value is inside it:

$$\underbrace{\hat{p}}_{\text{your estimate}} \pm \underbrace{E}_{\text{margin of error}}$$

Earlier lessons did this for a **mean** (an average of numbers). This lesson does the same thing for a **proportion** (a percentage of yes answers).

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## The symbols

The letters are the most confusing part of this lesson, so here they all are in one place:

| Symbol | Name | Meaning | Laptop example |
|---|---|---|---|
| $n$ | sample size | how many people you asked | 800 |
| $x$ | number of successes | how many said **yes** | 584 |
| $p$ | population proportion | the **true** fraction of everyone. Unknown, and it's what we're trying to estimate | ? |
| $\hat{p}$ ("p-hat"), also written $p'$ ("p-prime") | sample proportion | the fraction of **your sample** that said yes: $\hat{p} = x/n$ | $584/800 = 0.73$ |
| $\hat{q}$ ("q-hat") | proportion of failures | the fraction of your sample that said **no**: $\hat{q} = 1 - \hat{p}$ | $0.27$ |
| CL | confidence level | how sure you want to be, as a decimal | 95% → 0.95 |
| $z$ | critical value | how many standard errors wide the interval is; depends only on CL | 1.96 |
| $E$ or EBP | margin of error, or "error bound for a proportion" | how far the interval reaches on each side of $\hat{p}$ | 0.0308 |

> **$p$ vs $\hat{p}$:** $p$ (no hat) is the true answer for the whole population, which you never actually know. $\hat{p}$ (with hat) is what your sample gave you. The confidence interval is a range of believable values for $p$, built around $\hat{p}$.

$\hat{p}$ and $\hat{q}$ always add up to 1. If 73% said yes, then 27% said no.

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## Formulas

| Quantity | Formula |
|---|---|
| Sample proportion | $\hat{p} = \dfrac{x}{n}$ |
| Proportion of failures | $\hat{q} = 1 - \hat{p}$ |
| Check: is the normal distribution OK? | $n\hat{p}\hat{q} \geq 10$ |
| Area to the left of $z$ | $\dfrac{1 + \mathrm{CL}}{2}$ |
| Margin of error | $E = z\sqrt{\dfrac{\hat{p}\,\hat{q}}{n}}$ |
| Confidence interval | $\hat{p} - E < p < \hat{p} + E$ |
| Sample size needed | $n = \dfrac{z^2\,\hat{p}\,\hat{q}}{E^2}$, always rounded **up** |

Common $z$ values:

| Confidence level | Area to the left | $z$ |
|---:|---:|---:|
| 90% | 0.95 | 1.645 |
| 95% | 0.975 | 1.96 |
| 98% | 0.99 | 2.33 |
| 99% | 0.995 | 2.575 |

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## The recipe (do these steps in order)

1. **Write down $n$ and $x$** from the problem.
2. **Find $\hat{p} = x/n$ and $\hat{q} = 1 - \hat{p}$.**
3. **Check $n\hat{p}\hat{q} \geq 10$.** If it is, the normal distribution is a good approximation and you can use $z$.
4. **Find $z$** from the confidence level (see the next section).
5. **Find the margin of error $E = z\sqrt{\hat{p}\hat{q}/n}$.**
6. **Build the interval:** lower end $\hat{p} - E$, upper end $\hat{p} + E$.
7. **Write a sentence** saying what it means.

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## Step 3 in detail - why check $n\hat{p}\hat{q} \geq 10$?

Each person gives one of two answers (yes or no), so the number of yeses follows a **binomial distribution**. The confidence interval formula uses the **normal** (bell) curve instead, because $z$ values come from the normal curve.

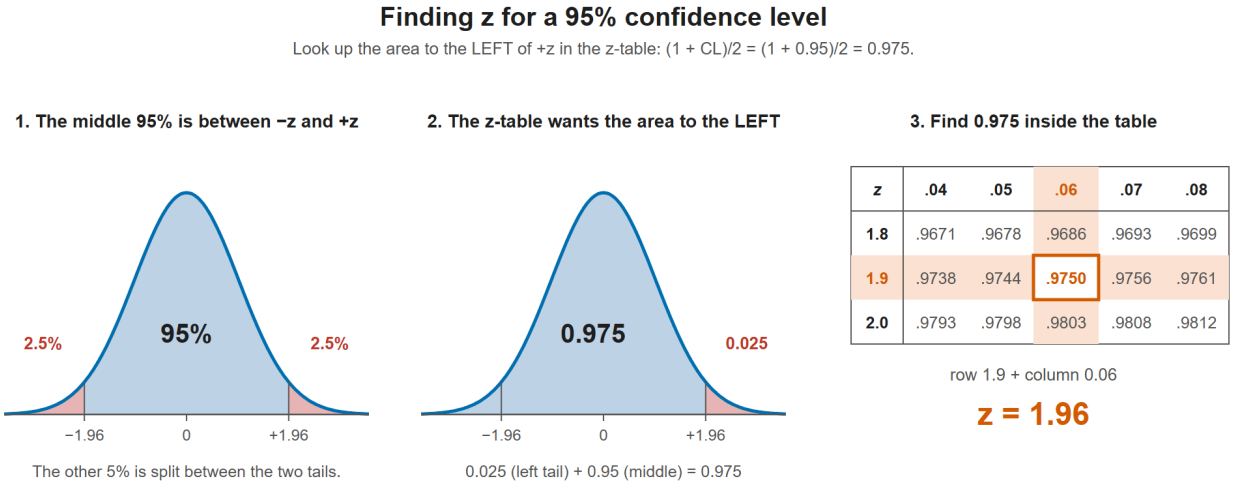
That's only allowed when the binomial distribution looks enough like a bell curve, which happens when the sample is big enough and $\hat{p}$ isn't too close to 0 or 1. The rule of thumb $n\hat{p}\hat{q} \geq 10$ tests this.

> Many textbooks use a slightly different check: $n\hat{p} \geq 10$ **and** $n\hat{q} \geq 10$, meaning at least 10 yeses and at least 10 noes. Use whichever your course uses.

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## Step 4 in detail - finding $z$ from the confidence level

**The picture.** A 95% confidence level means the middle 95% of the bell curve, between $-z$ and $+z$. The other 5% is split equally between the two tails: 2.5% on the left and 2.5% on the right.




The three steps for 95%. Left: the confidence level is the middle area, and the leftover 5% is split into two 2.5% tails. Middle: the z-table measures from the far left, so the area it wants also includes the left tail, 0.975 in total. Right: 0.9750 sits in row 1.9 and column 0.06, so z = 1.96.

A z-table gives the area to the **left** of a z-score. The area to the left of $+z$ is the middle 95% plus the left tail 2.5%:

$$\text{CL} + \frac{1 - \text{CL}}{2} = 0.95 + 0.025 = 0.975$$

**The shortcut formula** gives the same answer without the picture:

$$\text{area to the left} = \frac{1 + \text{CL}}{2} = \frac{1 + 0.95}{2} = \frac{1.95}{2} = 0.975$$

**Looking it up in the table.** Find 0.975 **inside** the positive z-table, then read off its row and column:

- Row: 1.9
- Column: 0.06
- So $z = 1.9 + 0.06 = 1.96$

**If the exact area isn't in the table**, take the two closest entries and average their z-scores. For example, for 90% the area to the left is 0.95. The table has 0.9495 (at $z = 1.64$) and 0.9505 (at $z = 1.65$). 0.95 is exactly halfway, so

$$z = \frac{1.64 + 1.65}{2} = 1.645$$

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## Step 5 in detail - what the margin of error formula means

$$E = z \times \underbrace{\sqrt{\frac{\hat{p}\,\hat{q}}{n}}}_{\text{standard error}}$$

- The square-root part is the **standard error**: roughly how much $\hat{p}$ would vary if you repeated the survey with different random people.
- $z$ sets how many standard errors you allow on each side. More confidence means a bigger $z$ and a wider interval.
- A bigger $n$ makes the standard error smaller, because larger samples are more reliable. That gives a narrower interval.

For why $\hat{p}\hat{q}$ appears here, see "The logic behind the formulas" in Lesson 9.5.

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## Example 1 - laptop ownership

800 randomly selected college students at university XYZ were asked if they own a laptop. 584 said yes.

### (a) Can the binomial distribution be approximated by a normal distribution?

**Write down what you know.** $n = 800$ and $x = 584$. The true proportion $p$ (all students at the university) is unknown, so we estimate it with $\hat{p}$.

**Find $\hat{p}$ and $\hat{q}$.**

$$\hat{p} = \frac{x}{n} = \frac{584}{800} = 0.73 \qquad \hat{q} = 1 - \hat{p} = 1 - 0.73 = 0.27$$

So 73% of the sample said yes and 27% said no.

**Check the condition.**

$$n\hat{p}\hat{q} = 800(0.73)(0.27) = 157.68 \geq 10 \checkmark$$

**Yes**, the normal distribution can be used.

### (b) 95% confidence interval and margin of error

**Find $z$.** The area to the left is $(1 + 0.95)/2 = 0.975$, which gives $z = 1.96$.

**Find the margin of error.** Work from the inside out:

$$\frac{\hat{p}\,\hat{q}}{n} = \frac{0.73 \times 0.27}{800} = \frac{0.1971}{800} = 2.46375 \times 10^{-4}$$

$$\sqrt{\frac{\hat{p}\,\hat{q}}{n}} = \sqrt{2.46375 \times 10^{-4}} = 0.015696$$

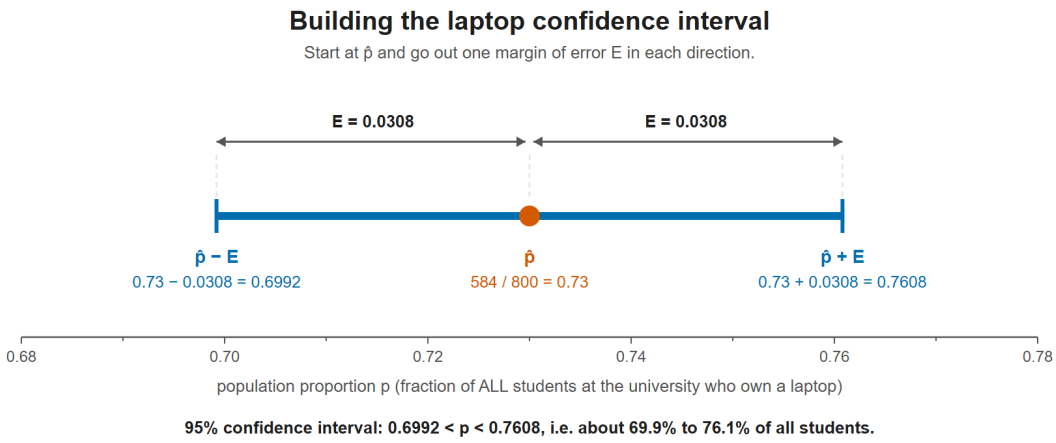
$$E = z\sqrt{\frac{\hat{p}\,\hat{q}}{n}} = 1.96 \times 0.015696 = 0.0307648 \approx 0.0308$$

**Build the interval.**

$$\text{lower} = \hat{p} - E = 0.73 - 0.0308 = 0.6992$$

$$\text{upper} = \hat{p} + E = 0.73 + 0.0308 = 0.7608$$

$$\boxed{0.6992 < p < 0.7608}$$




The interval is built outward from the sample result. The dot is $\hat{p}$ = 0.73, and each arm is one margin of error, so the two ends are $\hat{p} - E = 0.6992$ and $\hat{p} + E = 0.7608$.

**Interpretation.** We are 95% confident that between about **69.9% and 76.1%** of **all** students at university XYZ own a laptop.

The margin of error is 0.0308, about **±3.1 percentage points**.

## (c) What sample size gives a 2% margin of error at 90% confidence?

This time we're **not** building an interval. We're asking how many people we would need to survey.

**What changes.** $\hat{p} = 0.73$ and $\hat{q} = 0.27$ stay the same. The margin of error is now $E = 2\% = 0.02$, the confidence level is 90%, and $n$ is the unknown.

**Solve the margin of error formula for $n$.**

$$E = z\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

Square both sides. The square and the square root cancel:

$$E^2 = \frac{z^2\,\hat{p}\hat{q}}{n}$$

Multiply both sides by $n$:

$$nE^2 = z^2\,\hat{p}\hat{q}$$

Divide both sides by $E^2$:

$$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2}$$

**Find $z$ for 90%.** The area to the left is $(1 + 0.90)/2 = 0.95$, which gives $z = 1.645$. This is the "average two table values" case from Step 4.

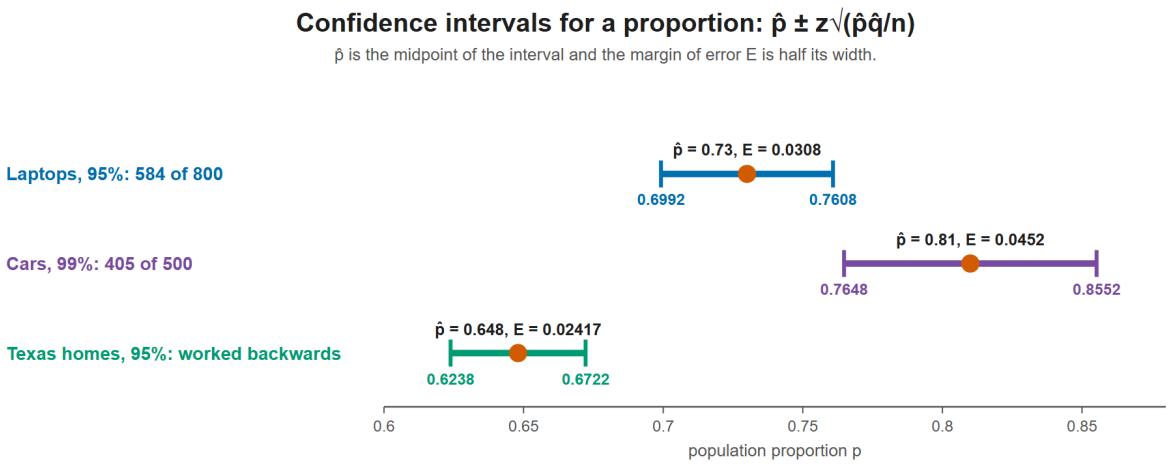
**Plug in.** Don't forget to square **both** $z$ and $E$:

$$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2} = \frac{(1.645)^2(0.73)(0.27)}{(0.02)^2} = \frac{2.706025 \times 0.1971}{0.0004} = \frac{0.53336}{0.0004} \approx 1333.39$$

**Round up**, even though normal rounding would give 1333:

$$n = 1334 \text{ students}$$

> **Why always round up?** With 1333 students the margin of error would be slightly **more** than 2%. We need **at least** enough people to reach 2%, so we go up to the next whole person.




Texas: p̂ = (0.62383 + 0.67217)/2 = 0.648 and E = (0.67217 − 0.62383)/2 = 0.02417, then n = z²p̂q̂/E² ≈ 1,500.

The three intervals from the lesson. The dot is p̂, the midpoint, and each arm is the margin of error E. The 99% car interval is the widest, because higher confidence uses a larger z.

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## Example 2 - car ownership

500 residents in town ABC were asked if they own a car. 405 said yes.

**Write down what you know and find $\hat{p}$, $\hat{q}$.**

$$n = 500, \quad x = 405, \quad \hat{p} = \frac{x}{n} = \frac{405}{500} = 0.81, \quad \hat{q} = 1 - \hat{p} = 1 - 0.81 = 0.19$$

So 81% of the sample own a car.

### (a) 99% confidence interval

**Find $z$.** The area to the left is $(1 + 0.99)/2 = 1.99/2 = 0.995$. In the table, 0.995 falls between $z = 2.57$ and $z = 2.58$, so average them:

$$z = \frac{2.57 + 2.58}{2} = 2.575$$

**Find the margin of error.**

$$\frac{\hat{p}\,\hat{q}}{n} = \frac{0.81 \times 0.19}{500} = \frac{0.1539}{500} = 0.0003078 \qquad \sqrt{\frac{\hat{p}\,\hat{q}}{n}} = \sqrt{0.0003078} = 0.01754$$

$$E = z\sqrt{\frac{\hat{p}\,\hat{q}}{n}} = 2.575 \times 0.01754 \approx 0.0452$$

**Build the interval.**

$$\hat{p} - E = 0.81 - 0.0452 = 0.7648 \qquad \hat{p} + E = 0.81 + 0.0452 = 0.8552$$

$$\boxed{0.7648 < p < 0.8552}$$

**Interpretation.** We are 99% confident that between about **76.5% and 85.5%** of all residents of town ABC own a car.

> Compare with Example 1: the margin of error here (±4.5 points) is wider than the laptop one (±3.1 points). Partly that's because 99% confidence uses a bigger $z$ (2.575 vs 1.96), and partly because the sample is smaller (500 vs 800).

### (b) What sample size gives a 3% margin of error at 98% confidence?

**Find $z$.** The area to the left is $(1 + 0.98)/2 = 1.98/2 = 0.99$. The closest table entry is 0.9901, at $z = 2.33$.

**Plug in**, with $E = 3\% = 0.03$:

$$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2} = \frac{(2.33)^2(0.81)(0.19)}{(0.03)^2} = \frac{5.4289 \times 0.1539}{0.0009} = \frac{0.83551}{0.0009} \approx 928.34$$

**Round up:** $n = 929$ residents.

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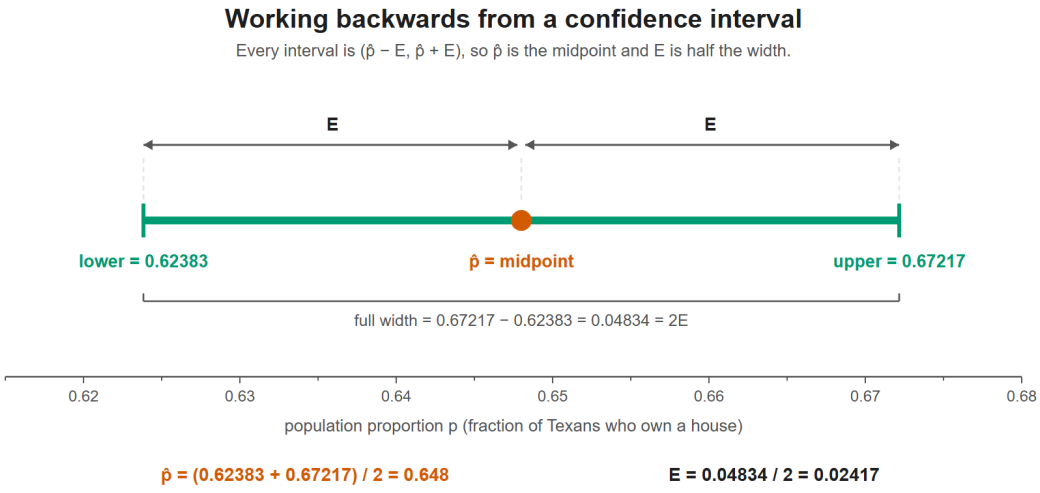
## Example 3 - working backwards from an interval

Some number of people in Texas were asked if they own a house. The 95% confidence interval for the true proportion of homeowners was $(0.62383,\ 0.67217)$.

This time the interval is **given**, and we work back to find the pieces that built it. The key fact is that every confidence interval has the shape

$$(\hat{p} - E,\ \ \hat{p} + E)$$

so $\hat{p}$ sits **exactly in the middle**, and $E$ is the distance from the middle to either end.




Only the two ends are given. Because the interval is p̂ ± E, p̂ is the point halfway between them (0.648), and E is half of the full width (0.04834 / 2 = 0.02417).

### (a) What is the sample proportion?

$\hat{p}$ is the **midpoint**, the average of the two ends:

$$\hat{p} = \frac{\text{lower} + \text{upper}}{2} = \frac{0.62383 + 0.67217}{2} = \frac{1.296}{2} = 0.648$$

### (b) What is the error bound?

**Method 1: upper end minus the middle.**

$$E = \text{upper} - \hat{p} = 0.67217 - 0.648 = 0.02417$$

**Method 2: half the width of the interval.** The full width is 2E, so

$$E = \frac{\text{upper} - \text{lower}}{2} = \frac{0.67217 - 0.62383}{2} = \frac{0.04834}{2} = 0.02417$$

Both methods give the same answer.

### (c) How many people were surveyed?

That's $n$, so use the sample-size formula. At 95% confidence $z = 1.96$, and $\hat{q} = 1 - \hat{p} = 1 - 0.648 = 0.352$:

$$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2} = \frac{(1.96)^2(0.648)(0.352)}{(0.02417)^2} = \frac{3.8416 \times 0.228096}{0.000584189} = \frac{0.87625}{0.000584189} \approx 1499.95$$

So about **1500 people** were surveyed.

> Here 1499.95 is just rounding noise in the given interval, so the real answer is 1500. Rounding up and rounding to the nearest whole number both give 1500.

### (d) How many said yes?

$\hat{p}$ is the **fraction** who said yes, and $x$ is the **number** who said yes. Since $\hat{p} = x/n$, multiply both sides by $n$:

$$x = n\,\hat{p} = 1500 \times 0.648 = 972$$

So **972 people** said yes, which is 64.8% of 1500.

---

## Common mistakes

- **Mixing up $p$ and $\hat{p}$.** $\hat{p}$ comes from the sample, and you calculate it. $p$ is the true population value, and you never know it exactly. The interval is a guess for $p$.
- **Forgetting to convert percentages.** A 2% margin of error is $E = 0.02$, not 2. A 95% confidence level is 0.95.
- **Looking up the confidence level directly in the z-table.** Look up the **area to the left**, $(1 + \text{CL})/2$. For 95%, look up 0.975, not 0.95.
- **Forgetting to square** $z$ or $E$ in the sample-size formula.
- **Rounding $n$ down.** Sample sizes always round **up**.
- **Misreading what "95% confident" means.** It does **not** mean there's a 95% chance that $p$ is in this particular interval. $p$ is a fixed number, and it either is or isn't inside. It means the **method** works 95% of the time: if you repeated the survey many times and built an interval each time, about 95% of those intervals would contain the true $p$.

---

## Key takeaways

- $\hat{p} = x/n$ is your estimate, and $\hat{q} = 1 - \hat{p}$. Check $n\hat{p}\hat{q} \geq 10$ before using the normal curve.
- Find $z$ by looking up the area to the left, $(1 + \text{CL})/2$, in the z-table.
- Confidence interval: $\hat{p} \pm z\sqrt{\hat{p}\hat{q}/n}$.
- Higher confidence means a bigger $z$ and a wider interval. A bigger sample means a narrower interval.

- Given an interval: its midpoint is $\hat{p}$, and half its width is $E$.
- Sample size: $n = z^2\hat{p}\hat{q}/E^2$, always rounded **up**. Lesson 9.5 covers sample size in more depth.

# Lesson 9.5 - How to Calculate the Sample Size

> **Video:** How To Calculate The Sample Size Given The Confidence Level & Margin of Error by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

## Formulas

**For a mean.** Start from the margin of error:

$$E = z\,\frac{\sigma}{\sqrt{n}} \xrightarrow{\text{square both sides}} E^2 = \frac{z^2\sigma^2}{n} \;\Rightarrow\; \boxed{n = \frac{z^2\sigma^2}{E^2}}$$

**For a proportion.** Same algebra, starting from the proportion's margin of error (Lesson 9.4):

$$E = z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \xrightarrow{\text{square both sides}} E^2 = \frac{z^2\,\hat{p}(1-\hat{p})}{n} \;\Rightarrow\; \boxed{n = \frac{z^2\,\hat{p}\,(1-\hat{p})}{E^2}}$$

It is the mean formula with $\sigma^2$ replaced by $\hat{p}(1-\hat{p})$, as explained below.

If $\hat{p}$ is unknown, use $\hat{p} = 0.5$. That makes $\hat{p}(1-\hat{p})$ as large as possible, so the resulting $n$ is big enough whatever the true proportion is.

Always **round** $n$ **up** to the next whole number.

---

## The logic behind the formulas

**1. Every sample gives a different answer.** Survey $n$ people and you get a sample proportion $\hat{p}$. A different random group would give a slightly different $\hat{p}$. Across many samples, the $\hat{p}$ values form a bell curve centred on the true $p$. The spread of that curve is the **standard error**:

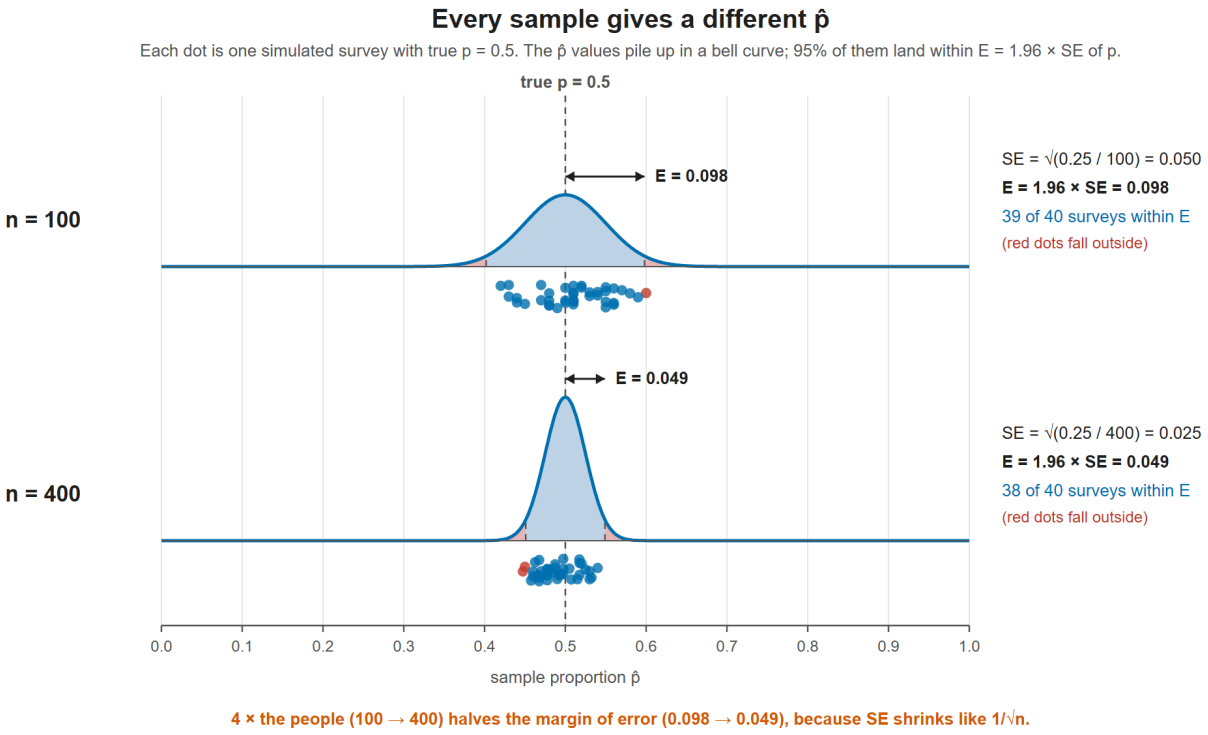
$$SE = \sqrt{\frac{p(1-p)}{n}}$$

**Why $p$ here and not $\hat{p}$?** The SE describes how much $\hat{p}$ wobbles from sample to sample. That wobble comes from the population the samples are drawn from, so the exact formula uses the true $p$. But $p$ is exactly what we don't know. In practice we plug in the best stand-in available: the sample's $\hat{p}$ when building an interval from data (Lesson 9.4), or a guess, or 0.5, when planning a survey before any data exists (point 6). That swap is why the formulas at the top are written with $\hat{p}$. With a decent-sized sample $\hat{p}$ is close to $p$, so the estimated SE is close to the true one.

**2. The margin of error is $z$ standard errors.** On a bell curve, 95% of values lie within 1.96 standard deviations of the centre. So 95% of the time, $\hat{p}$ lands within $1.96 \times SE$ of the truth:

$$E = z \times SE$$

$E$ is how far off you are willing to be, and $z$ is how sure you want to be of staying within that distance.



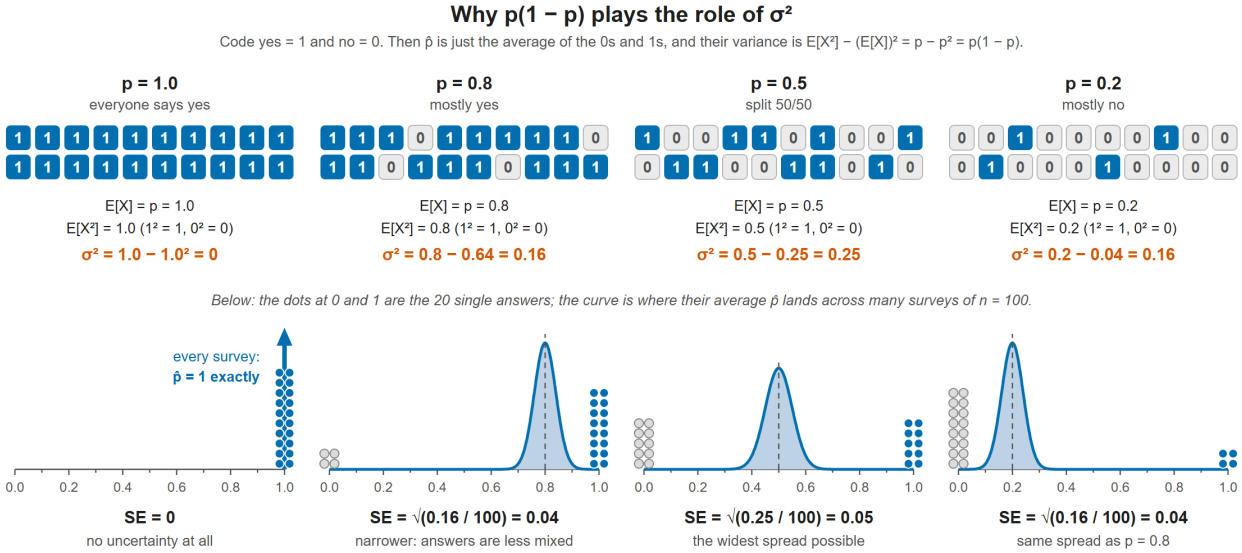

Each dot is one simulated survey. Its $\hat{p}$ misses the true $p = 0.5$ by a different amount each time, but about 95% of the dots land inside $\pm E$. The bottom row previews point 4: four times as many people makes the curve half as wide.

**3. Why $p(1-p)$ plays the role of $\sigma^2$.** Code each answer as 1 (yes) or 0 (no). Then $\hat{p}$ is just the average of those 0s and 1s, and their variance is

$$\sigma^2 = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$$

since $1^2 = 1$ and $0^2 = 0$. Intuitively:

- If everyone says yes ($p = 1$), every sample looks the same: $p(1-p) = 0$, so there is no uncertainty.
- If the population is split 50/50, samples can easily tilt either way, so the variation is largest: $p(1-p) = 0.25$ .

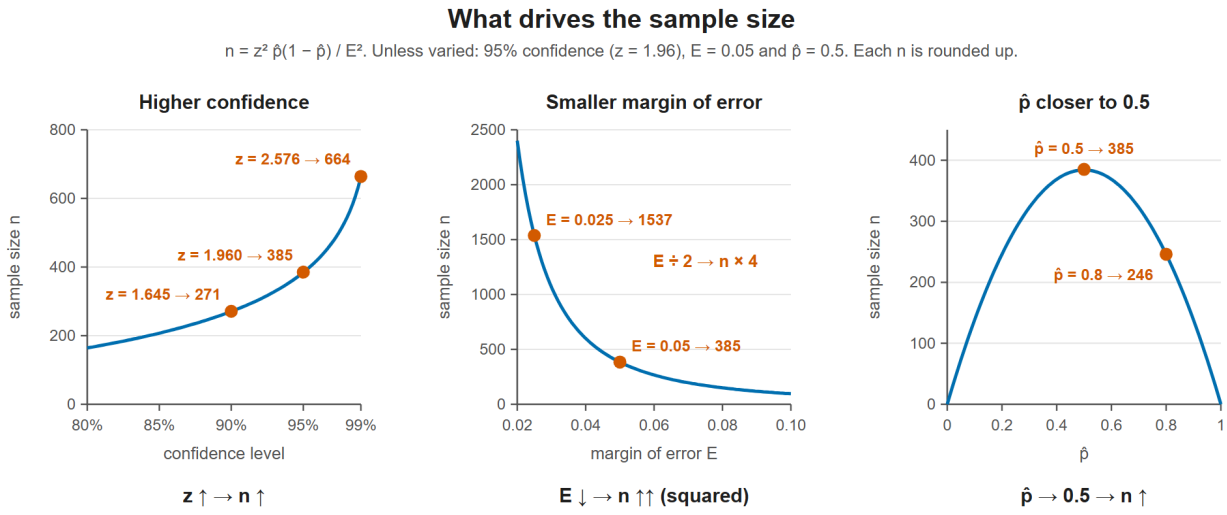


A single answer can only be 0 or 1 (the dots), but the average of 100 answers clusters tightly around $p$ (the curve). The more mixed the 0s and 1s, the bigger $p(1-p)$, and the wider $\hat{p}$ spreads from survey to survey. $p = 0.2$ and $p = 0.8$ give the same spread, since $0.2 \times 0.8 = 0.8 \times 0.2$: only how mixed the answers are matters, not which side is the majority.

**4. Why divide by $n$.** In a small sample, a few unusual answers move $\hat{p}$ a lot. In a large one they get diluted. The variance shrinks like $1/n$, so $SE$ shrinks like $1/\sqrt{n}$. Because of the square root, **halving** the margin of error takes **4 times** as many people.

**5. Turn it around.** Normally you collect data and then compute $E$. Sample-size planning runs the other way: you choose the $E$ you want and solve for the $n$ that achieves it.

| Change | Effect on $n$ | Why |
|---|---|---|
| Higher confidence ($z$ ↑) | $n$ ↑ | Being more sure needs more evidence |
| Smaller margin of error ($E$ ↓) | $n$ ↑↑ (squared) | More precision needs more data |
| $p$ closer to 0.5 | $n$ ↑ | A more mixed population is less predictable |




Each panel changes one ingredient and holds the others at 95%, $E = 0.05$, $\hat{p} = 0.5$. The middle panel is the steepest: halving $E$ multiplies $n$ by 4.

**6. Why $\hat{p} = 0.5$ is the safe choice.** You plan the survey before you have data, so $p$ is unknown. The safe move is to use the $\hat{p}$ that makes $n$ as large as possible. Since $z$ and $E$ are fixed, that means finding the maximum of

$$f(\hat{p}) = \hat{p}(1-\hat{p}) = \hat{p} - \hat{p}^2, \qquad 0 \le \hat{p} \le 1$$

*With calculus.* Set the first derivative to zero:

$$f'(\hat{p}) = 1 - 2\hat{p} = 0 \;\Rightarrow\; \hat{p} = 0.5$$

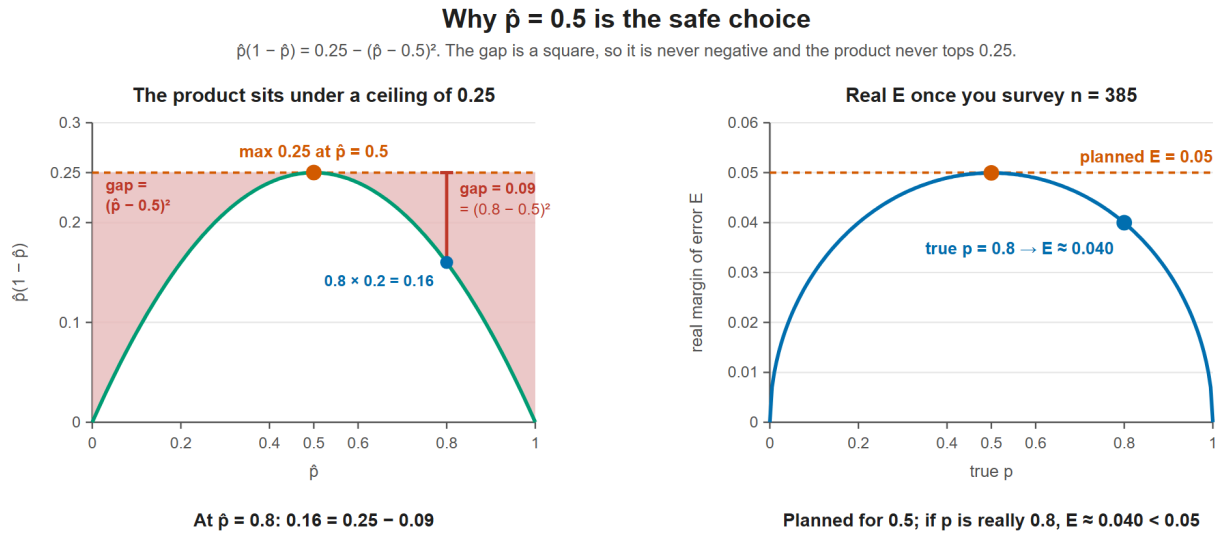
The second derivative is $f''(\hat{p}) = -2 < 0$, so the curve bends downward everywhere and $\hat{p} = 0.5$ is a maximum, not a minimum. Its value is $f(0.5) = 0.5 \times 0.5 = 0.25$. At the ends, $f(0) = f(1) = 0$, so 0.25 is the largest value on the whole range $0 \le \hat{p} \le 1$.

*Without calculus.* Completing the square gives the same answer:

$$\hat{p}(1-\hat{p}) = 0.25 - (\hat{p} - 0.5)^2$$

The squared term is never negative, so the product can never exceed 0.25, which happens only at $\hat{p} = 0.5$.

Planning for this worst case guarantees $n$ is large enough. If the true $p$ turns out to be, say, 0.8, the real margin of error simply comes out a bit smaller than planned.




Left: the shaded gap between the curve and 0.25 is $(\hat{p} - 0.5)^2$, which is zero only at $\hat{p} = 0.5$. Right: survey the planned $n = 385$, and if $p$ turns out to be 0.8 the margin of error is $1.96\sqrt{0.16/385} \approx 0.040$, comfortably inside the 0.05 you asked for.

> **In one sentence:** sample answers wobble around the truth. The wobble grows with how mixed the population is and shrinks with $\sqrt{n}$, so you pick $n$ large enough that $z$ times the wobble fits inside the error you'll accept.

---

## Example 1 - working backwards to the class size

The mean test score was 84 with standard deviation $\sigma = 15$. The margin of error at 95% confidence was $E = 5.367$. How many students were sampled?

$$n = \frac{z^2\sigma^2}{E^2} = \frac{(1.96)^2(15)^2}{(5.367)^2} \approx 30.0 \;\Rightarrow\; n = 30 \text{ students}$$

(The mean, 84, isn't needed.)

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## Example 2 - planning a survey on a proportion

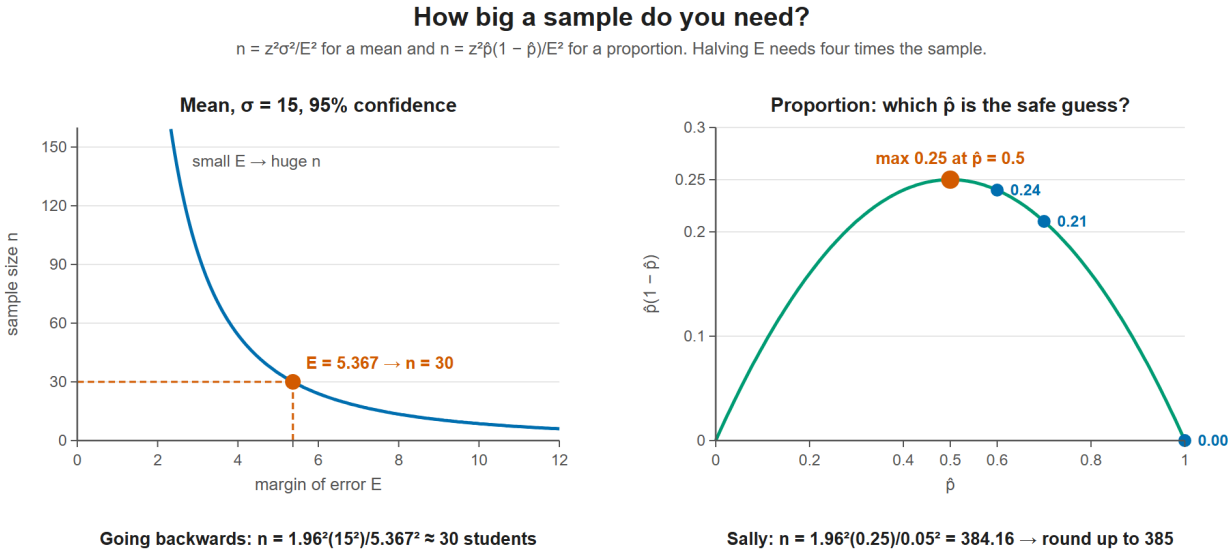
Sally wants to estimate the proportion of high-school students planning to go to college, with **95%** confidence and a **5%** margin of error ($E = 0.05$). She has no prior estimate of $\hat{p}$.

**Which $\hat{p}$ makes $\hat{p}(1-\hat{p})$ largest?**

| $\hat{p}$ | $1-\hat{p}$ | Product |
|---|---|---|
| 1.0 | 0.0 | 0.00 |
| 0.7 | 0.3 | 0.21 |
| 0.6 | 0.4 | 0.24 |
| **0.5** | **0.5** | **0.25** (largest) |

$$n = \frac{z^2\,\hat{p}(1-\hat{p})}{E^2} = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2} = \frac{0.9604}{0.0025} = 384.16 \;\Rightarrow\; n = 385$$

Round **up**: 384 students would not quite reach the 5% margin.




Left: the required n grows fast as the margin of error shrinks, and Example 1's E = 5.367 corresponds to n = 30. Right: $\hat{p}(1-\hat{p})$ is largest at $\hat{p} = 0.5$, so using 0.5 when nothing is known gives the safest (largest) sample size.

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## Key takeaways

- Mean: $n = z^2\sigma^2/E^2$. Proportion: $n = z^2\hat{p}(1-\hat{p})/E^2$.
- Both come from solving $E = z \times SE$ for $n$. For yes/no data the variance is $p(1-p)$, which takes the place of $\sigma^2$.
- With no estimate of $\hat{p}$, use 0.5. This gives the largest, safest $n$.
- Always round **up**.
- A smaller margin of error or a higher confidence level requires a larger sample.

# Lesson 9.6 - Chebyshev's Theorem

> **Video:** Chebyshev's Theorem by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

## Statement

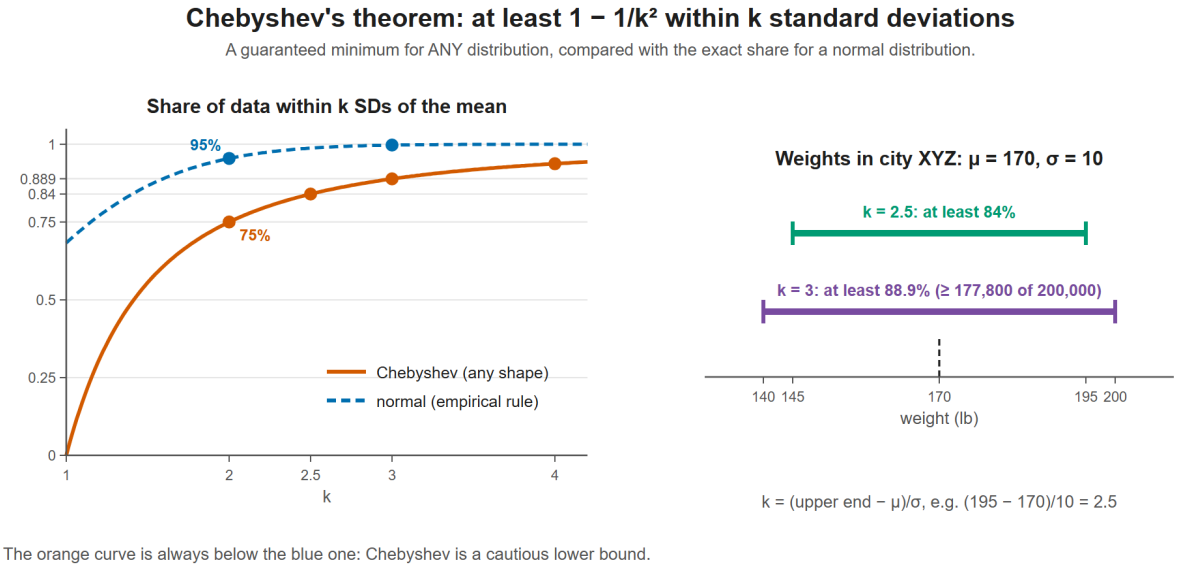
For **any** distribution, whatever its shape, the proportion of data within $k$ standard deviations of the mean is **at least**

$$1 - \frac{1}{k^2}, \qquad k > 1$$

- It gives a **minimum**, not an exact value. The true proportion can be higher.
- It works for **all** distributions, not only normal ones.
- In the video the formula is spoken as "1 over 1 minus k squared". The correct form is $1 - 1/k^2$.

## Chebyshev vs the empirical rule

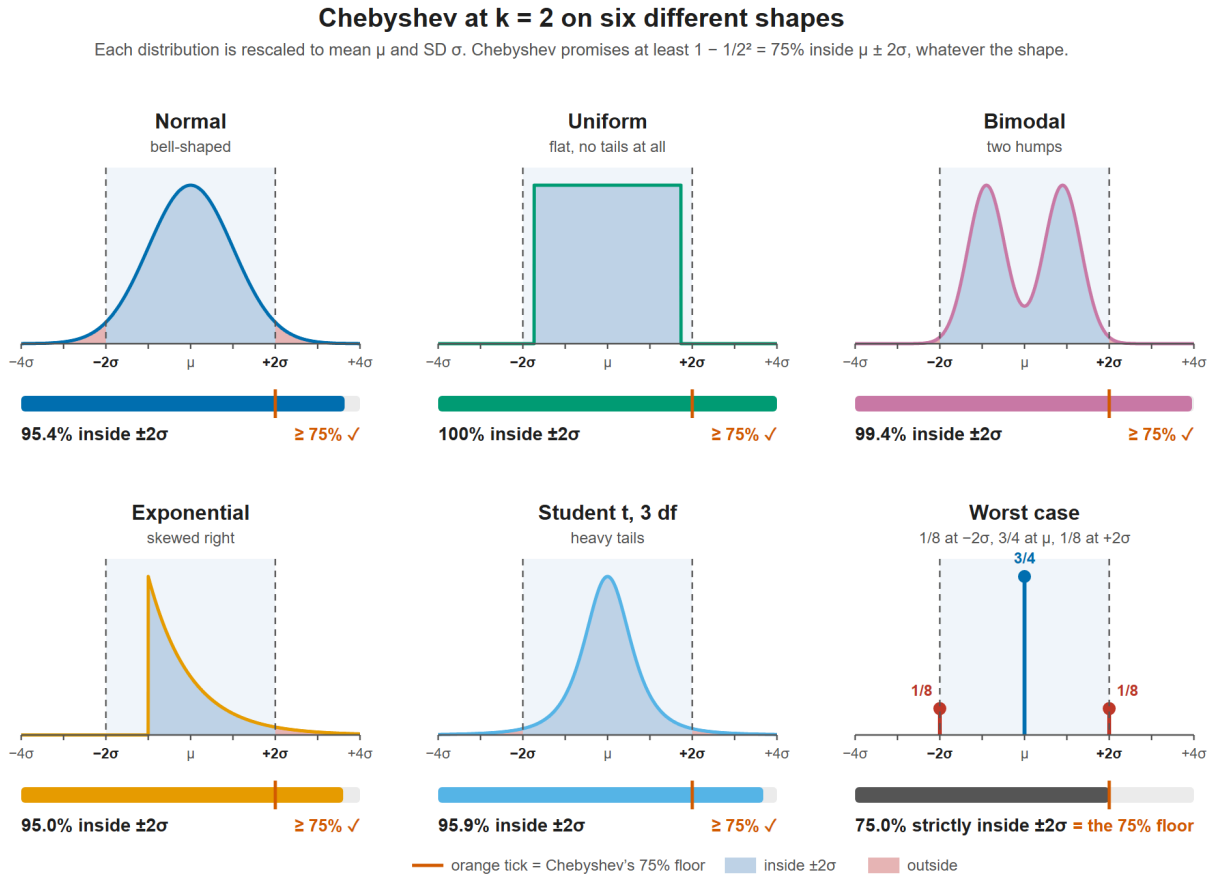
| Within | Empirical rule (normal only) | Chebyshev (any distribution, minimum) |
|---|---|---|
| $1\sigma$ | 68% | — ($k$ must be $> 1$) |
| $2\sigma$ | 95% | $1 - \frac{1}{4} = 75\%$ |
| $3\sigma$ | 99.7% | $1 - \frac{1}{9} \approx 88.9\%$ |
| $4\sigma$ | ≈ 99.99% | $1 - \frac{1}{16} = 93.75\%$ |




Left: Chebyshev's guaranteed minimum (orange) is always below the exact share for normal data (blue). For example, it guarantees 75% within 2 SDs, while a normal distribution has 95%. Right: the two intervals from the weight example.

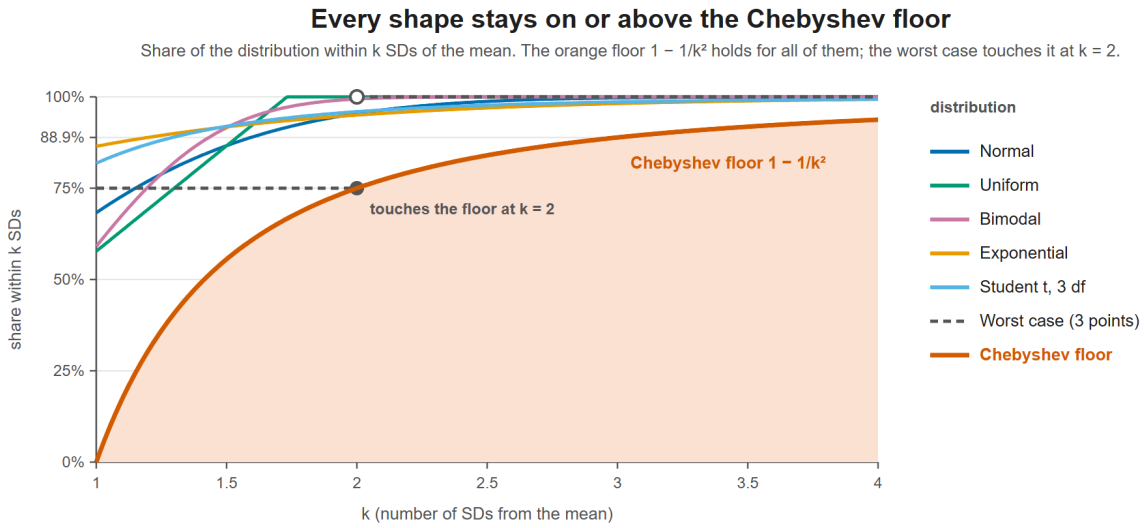
## Chebyshev on other famous distributions

"Any distribution" is a strong claim, so here it is tested on six very different shapes. Each one is rescaled to the same mean $\mu$ and standard deviation $\sigma$, so the axis reads directly in SDs from the mean.




| Share within $k\sigma$ of $\mu$ | $k = 1.5$ | $k = 2$ | $k = 3$ |
|---|---|---|---|
| Normal | 86.6% | 95.4% | 99.7% |
| Uniform | 86.6% | 100% | 100% |
| Bimodal (two humps) | 91.6% | 99.4% | ≈ 100% |
| Exponential (skewed) | 91.8% | 95.0% | 98.2% |
| Student $t$, 3 df (heavy tails) | 92.0% | 95.9% | 98.6% |
| Worst case (three points) | 75% | 75% | 100% |
| **Chebyshev minimum** $1 - 1/k^2$ | **55.6%** | **75%** | **88.9%** |

- **Every row is at or above the Chebyshev row.** Very different shapes all satisfy the guarantee.
- **Real shapes beat the bound by a lot.** At $k = 2$, five of the six have 95% or more, far above the promised 75%.
- **The bound can't be raised, though.** The worst case puts $\frac{1}{8}$ of the data at $\mu - 2\sigma$, $\frac{3}{4}$ at $\mu$ and $\frac{1}{8}$ at $\mu + 2\sigma$. Its SD is exactly $\sigma$, since $\sqrt{\frac{1}{8}(2\sigma)^2 + \frac{1}{8}(2\sigma)^2} = \sigma$, yet only 75% lies strictly inside $\mu \pm 2\sigma$. Because one shape lands exactly on 75%, no rule that covers all shapes can promise more.




The orange curve is the floor $1 - 1/k^2$. Each coloured line shows the true share within $k$ SDs for one shape, and none dips below the floor. The dashed worst case touches it at $k = 2$, which is why Chebyshev's theorem is sometimes called "sharp".

**What "sharp" means.** A bound is *sharp* (or *tight*) when it can't be improved. Suppose someone proposed a stronger rule, such as "at least 80% of any data lies within $2\sigma$". The worst case breaks it, because it has only 75%. So 75% is the best guarantee that holds for every possible shape. Real data usually beats it easily, but no universal rule can promise more.

**A worst case for any $k$.** The example above touches the floor only at $k = 2$ because it was built for $k = 2$. The same recipe works for any $k > 1$:

| Value | $\mu - k\sigma$ | $\mu$ | $\mu + k\sigma$ |
|---|---|---|---|
| Share of the data | $\frac{1}{2k^2}$ | $1 - \frac{1}{k^2}$ | $\frac{1}{2k^2}$ |

- **Its mean is $\mu$**, because the two outer points balance each other.
- **Its SD is exactly $\sigma$.** Only the outer points are away from the mean, each by $k\sigma$:

$$\text{variance} = \frac{1}{2k^2}(k\sigma)^2 + \frac{1}{2k^2}(k\sigma)^2 = \frac{\sigma^2}{2} + \frac{\sigma^2}{2} = \sigma^2$$

- **Exactly $1/k^2$ of the data sits $k\sigma$ from the mean**, so the share strictly inside $\mu \pm k\sigma$ is exactly $1 - 1/k^2$, right on the floor.

With $k = 2$ this gives $\frac{1}{8}$, $\frac{3}{4}$, $\frac{1}{8}$, the worst case in the figures. With $k = 3$ it gives $\frac{1}{18}$ at each end and $\frac{8}{9}$ at $\mu$, which touches the floor at 88.9%. So the bound is sharp at every $k$; each worst-case example just touches the curve at its own $k$.

## Finding $k$

If an interval is centered on the mean, $\mu \pm k\sigma$, then

$$k = \frac{x - \mu}{\sigma}$$

using the upper end $x$ (the lower end gives the same $k$).

## Example - weights in city XYZ

Mean weight $\mu = 170$ lb, $\sigma = 10$ lb, distribution shape unknown.

**(a) Minimum percentage of people weighing between 145 and 195 lb**

Both ends are 25 lb from the mean, so

$$k = \frac{x - \mu}{\sigma} = \frac{195 - 170}{10} = 2.5$$

$$1 - \frac{1}{k^2} = 1 - \frac{1}{2.5^2} = 1 - \frac{1}{6.25} = 0.84$$

**At least 84%** of residents weigh between 145 and 195 lb. The real figure could be 86%, 89%, 95%, ...

**(b) The city has 200,000 residents. What is the minimum number weighing between 140 and 200 lb?**

Both ends are 30 lb from the mean, so

$$k = \frac{x - \mu}{\sigma} = \frac{200 - 170}{10} = 3, \qquad 1 - \frac{1}{k^2} = 1 - \frac{1}{9} = \frac{8}{9} \approx 0.889$$

$$\left(1 - \frac{1}{k^2}\right) \times N = \frac{8}{9} \times 200{,}000 \approx 177{,}778 \text{ people}$$

(The video rounds to 0.889 first and gets 177,800.)

So $P(140 \le X \le 200) \ge 0.889$, and **at least about 177,800** residents weigh between 140 and 200 lb.

---

## Key takeaways

- Chebyshev: at least $1 - 1/k^2$ of any data set lies within $k$ SDs of the mean ($k > 1$).
- Use it when the distribution is **unknown or not normal**. Use the empirical rule only for normal data.
- $k = (x - \mu)/\sigma$ for an interval centered on the mean.
- Answers are **lower bounds**: say "at least".